

	W	B	R	S	I	V	A	P	N
W	3	4	12	2	4	10	14	3	1
B	4	4	0	11	5	5	2	4	20
R	5	5	0	1	1	5	0	3	0
S	1	0	5	0	1	3	5	2	0
I	1	8	10	1	0	2	3	0	0
V	8	10	0	0	2	2	0	3	1
A	7	3	4	2	5	4	0	1	0
P	0	8	6	0	1	1	4	0	0
N	14	3	0	1	1	1	0	7	0

Table 2.3: Counting Digrams

We have already decided that $W = e$, but if we had extended the table to include low-frequency letters, we would see that W contacts many of these letters, too, which is another characteristic of e . This helps to confirm our guess.

The vowels a, i, o tend to avoid each other. If we look at the R row, we see that R does not precede S, I, A, N very often. But a look at the R column shows that R follows S, I, A fairly often. So we suspect that R is not one of a, i, o . V and N are out because they would require a, i , or o to precede $W = e$ quite often, which is unlikely. Continuing, we see that the most likely possibilities for a, i, o are S, I, P in some order.

The letter n has the property that around 80% of the letters that precede it are vowels. Since we already have identified W, S, I, P as vowels, we see that R and A are the most likely candidates. We'll have to wait to see which is correct.

The letter h often appears before e and rarely after it. This tells us that $N = h$.

The most common digram is th . Therefore, $B = t$.

Among the frequent letters, r and s remain, and they should equal V and one of A, R . Since r pairs more with vowels and s pairs more with consonants, we see that V must be s and r is represented by either A or R .

The combination rn should appear more than nr , and AR is more frequent than RA , so our guess is that $A = r$ and $R = n$.

We can continue the analysis and determine that $S = o$ (note that to is much more common than ot), $I = i$, and $P = a$ are the most likely choices. We have therefore determined reasonable guesses for 382 of the 520 characters in the text:

L W N S O Z B N W V W B A Y B N V B S
 e h o t h e s e l r t h s t o

Q W V W O H W D I Z W R B B N P B P ...
 e s e e i e n t t h a t a ...

At this point, knowledge of the language, middle-level frequencies ($l, d, \dots$), and educated guesses can be used to fill in the remaining letters. For example, in the first line a good guess is that $Y = u$ since then the word *truths* appears. Of course, there is a lot of guesswork, and various hypotheses need to be tested until one works.

Since the preceding should give the spirit of the method, we skip the remaining details. The decrypted message, with spaces (but not punctuation) added, is as follows (the text is from the middle of the Declaration of Independence):

we hold these truths to be self evident that all men are created equal that they are endowed by their creator with certain unalienable rights that among these are life liberty and the pursuit of happiness that to secure these rights governments are instituted among men deriving their just powers from the consent of the governed that whenever any form of government becomes destructive of these ends it is the right of the people to alter or to abolish it and to institute new government laying its foundation on such principles and organizing its powers in such form as to seem most likely to effect their safety and happiness

2.5 Sherlock Holmes

Cryptography has appeared in many places in literature, for example, in the works of Edgar Allen Poe (*The Gold Bug*), William Thackeray (*The History of Henry Esmond*), Jules Verne (*Voyage to the Center of the Earth*), and Agatha Christie (*The Four Suspects*).

Here we give a summary of an enjoyable tale by Arthur Conan Doyle, in which Sherlock Holmes displays his usual cleverness, this time by breaking a cipher system. We cannot do the story justice here, so we urge the reader to read *The Adventure of the Dancing Men* in its entirety. The following is a cryptic, and cryptographic, summary of the plot.

Mr. Hilton Cubitt, who has recently married the former Elsie Patrick, mails Sherlock Holmes a letter. In it is a piece of paper with dancing stick

figures that he found in his garden at Riding Thorpe Manor:



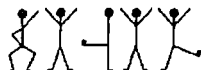
Two weeks later, Cubitt finds another series of figures written in chalk on his toolhouse door:



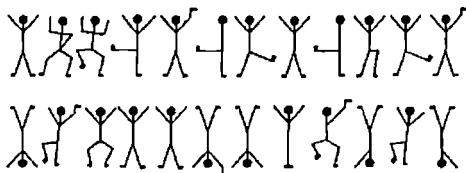
Two mornings later another sequence appears:



Three days later, another message appears:



Cubitt gives copies of all of these to Holmes, who spends the next two days making many calculations. Suddenly, Holmes jumps from his chair, clearly having made a breakthrough. He quickly sends a long telegram to someone and then waits, telling Watson that they will probably be going to visit Cubitt the next day. But two days pass with no reply to the telegram, and then a letter arrives from Cubitt with yet another message:



Holmes studies it and says they need to travel to Riding Thorpe Manor as soon as possible. A short time later, a reply to Holmes's telegram arrives, and Holmes indicates that the matter has become even more urgent. When Holmes and Watson arrive at Cubitt's house the next day, they find the police already there. Cubitt has been shot dead. His wife, Elsie, has also been shot and is in critical condition (although she survives). Holmes asks

several questions and then has someone deliver a note to a Mr. Abe Slaney at nearby Elrige's Farm. Holmes then explains to Watson and the police how he decrypted the messages. First, he guessed that the flags on some of the figures indicated the ends of words. He then noticed that the most common figure was



so it was likely *E*. This gave the fourth message as *-E-E-*. The possibilities *LEVER*, *NEVER*, *SEVER* came to mind, but since the message was probably a one word reply to a previous message, Holmes guessed it was *NEVER*. Next, Holmes observed that



had the form *E--E*, which could be *ELSIE*. The third message was therefore *---E ELSIE*. Holmes tried several combinations, finally settling on *COME ELSIE* as the only viable possibility. The first message therefore was *-M -ERE --E SL-NE-*. Holmes guessed that the first letter was *A* and the third letter as *H*, which gave the message as *AM HERE A-E SLANE-*. It was reasonable to complete this to *AM HERE ABE SLANEY*. The second message then was *A- ELRI-ES*. Of course, Holmes correctly guessed that this must be stating where Slaney was staying. The only letters that seemed reasonable completed the phrase to *AT ELRIGES*. It was after decrypting these two messages that Holmes sent a telegram to a friend at the New York Police Bureau, who sent back the reply that Abe Slaney was "the most dangerous crook in Chicago." When the final message arrived, Holmes decrypted it to *ELSIE -RE-ARE TO MEET THY GO-*. Since he recognized the missing letters as *P, P, D*, respectively, Holmes became very concerned and that's why he decided to make the trip to Riding Thorpe Manor.

When Holmes finishes this explanation, the police urge that they go to Elrige's and arrest Slaney immediately. However, Holmes suggests that is unnecessary and that Slaney will arrive shortly. Sure enough, Slaney soon appears and is handcuffed by the police. While waiting to be taken away, he confesses to the shooting (it was somewhat in self defense, he claims) and says that the writing was invented by Elsie Patrick's father for use by his gang, the Joint, in Chicago. Slaney was engaged to be married to Elsie, but she escaped from the world of gangsters and fled to London. Slaney finally

traced her location and sent the secret messages. But why did Slaney walk into the trap that Holmes set? Holmes shows the message he wrote:



From the letters already deduced, we see that this says *COME HERE AT ONCE*. Slaney was sure this message must have been from Elsie since he was certain no one outside of the Joint could write such messages. Therefore, he made the visit that led to his capture.

Comments

What Holmes did was solve a simple substitution cipher, though he did this with very little data. As with most such ciphers, both frequency analysis and a knowledge of the language are very useful. A little luck is nice, too, both in the form of lucky guesses and in the distribution of letters. Note how overwhelmingly *E* was the most common letter. In fact, it appeared 11 times among the 38 characters in the first four messages. This gave Holmes a good start. If Elsie had been Carol and Abe Slaney had been John Smith, the decryption would probably have been more difficult.

Authentication is an important issue in cryptography. If Eve breaks Alice's cryptosystem, then Eve can often masquerade as Alice in communications with Bob. Safeguards against this are important. The judges gave Abe Slaney many years to think about this issue.

The alert reader might have noticed that we cheated a little when decrypting the messages. The same symbol represents the *V* in *NEVER* and the *P*s in *PREPARE*. This is presumably due to a misprint and has occurred in every printed version of the work, starting with the story's first publication back in 1903. In the original text, the *R* in *NEVER* is written as the *B* in *ABE*, but this is corrected in later editions (however, in some later editions, the first *C* in the message Holmes wrote is given an extra arm and therefore looks like the *M*). If these mistakes had been in the text that Holmes was working with, he would have had a very difficult time decrypting and would have rightly concluded that the Joint needed to use error correction techniques in their transmissions. In fact, some type of error correction should be used in conjunction with almost every cryptographic protocol.

2.6 The Playfair and ADFGX Ciphers

The Playfair and ADFGX ciphers were used in World War I by the British and the Germans, respectively. By modern standards, they are fairly weak

systems, but they took real effort to break at the time.

The Playfair system was invented around 1854 by Sir Charles Wheatstone, who named it after his friend, the Baron Playfair of St. Andrews, who worked to convince the government to use it. In addition to being used in World War I, it was used by the British forces in the Boer War.

The key is a word, for example, *playfair*. The repeated letters are removed, to obtain *playfir*, and the remaining letters are used to start a 5×5 matrix. The remaining spaces in the matrix are filled in with the remaining letters in the alphabet, with *i* and *j* being treated as one letter:

<i>p</i>	<i>l</i>	<i>a</i>	<i>y</i>	<i>f</i>
<i>i</i>	<i>r</i>	<i>b</i>	<i>c</i>	<i>d</i>
<i>e</i>	<i>g</i>	<i>h</i>	<i>k</i>	<i>m</i>
<i>n</i>	<i>o</i>	<i>q</i>	<i>s</i>	<i>t</i>
<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>z</i>

Suppose the plaintext is *meet at the schoolhouse*. Remove spaces and divide the text into groups of two letters. If there is a doubled letter appearing as a group, insert an *x* and regroup. Add an extra *x* at the end to complete the last group, if necessary. Our plaintext becomes

me et at th es ch ox ol ho us ex.

Now use the matrix to encrypt each two letter group by the following scheme:

- If the two letters are not in the same row or column, replace each letter by the letter that is in its row and is in the column of the other letter. For example, *et* becomes *MN*, since *M* is in the same row as *e* and the same column as *t*, and *N* is in the same row as *t* and the same column as *e*.
- If the two letters are in the same row, replace each letter with the letter immediately to its right, with the matrix wrapping around from the last column to the first. For example, *me* becomes *EG*.
- If the two letters are in the same column, replace each letter with the letter immediately below it, with the matrix wrapping around from the last row to the first. For example, *ol* becomes *VR*.

The ciphertext in our example is

EG MN FQ QM KN BK SV VR GQ XN KU.

To decrypt, reverse the procedure.

The system succumbs to a frequency attack since the frequencies of the various digrams (two-letter combinations) in English have been tabulated.

Of course, we only have to look for the most common digrams; they should correspond to the most common digrams in English: *th*, *he*, *an*, *in*, *re*, *es*, Moreover, a slight modification yields results more quickly. For example, both of the digrams *re* and *er* are very common. If the pairs *IG* and *GI* are common in the ciphertext, then a good guess is that *e*, *i*, *r*, *g* form the corners of a rectangle in the matrix. Another weakness is that each plaintext letter has only five possible corresponding ciphertext letters. Also, unless the keyword is long, the last few rows of the matrix are predictable. Observations such as these allow the system to be broken with a ciphertext only attack. For more on its cryptanalysis, see [Gaines].

The ADFGX cipher proceeds as follows. Put the letters of the alphabet into a 5×5 matrix. The letters *i* and *j* are treated as one, and the columns of the matrix are labeled with the letters *A*, *D*, *F*, *G*, *X*. For example, the matrix could be

	<i>A</i>	<i>D</i>	<i>F</i>	<i>G</i>	<i>X</i>
<i>A</i>	<i>p</i>	<i>g</i>	<i>c</i>	<i>e</i>	<i>n</i>
<i>D</i>	<i>b</i>	<i>q</i>	<i>o</i>	<i>z</i>	<i>r</i>
<i>F</i>	<i>s</i>	<i>l</i>	<i>a</i>	<i>f</i>	<i>t</i>
<i>G</i>	<i>m</i>	<i>d</i>	<i>v</i>	<i>i</i>	<i>w</i>
<i>X</i>	<i>k</i>	<i>u</i>	<i>y</i>	<i>x</i>	<i>h</i>

Each plaintext letter is replaced by the label of its row and column. For example, *s* becomes *FA*, and *z* becomes *DG*. Suppose the plaintext is

Kaiser Wilhelm.

The result of this initial step is

XA FF GG FA AG DX GX GG FD XX AG FD GA.

So far, this is a disguised substitution cipher. The next step increases the complexity significantly. Choose a keyword, for example, *Rhein*. Label the columns of a matrix by the letters of the keyword and put the result of the initial step into another matrix:

<i>R</i>	<i>H</i>	<i>E</i>	<i>I</i>	<i>N</i>
<i>X</i>	<i>A</i>	<i>F</i>	<i>F</i>	<i>G</i>
<i>G</i>	<i>F</i>	<i>A</i>	<i>A</i>	<i>G</i>
<i>D</i>	<i>X</i>	<i>G</i>	<i>X</i>	<i>G</i>
<i>G</i>	<i>F</i>	<i>D</i>	<i>X</i>	<i>X</i>
<i>A</i>	<i>G</i>	<i>F</i>	<i>D</i>	<i>G</i>
<i>A</i>				

Now reorder the columns so that the column labels are in alphabetic order:

<i>E</i>	<i>H</i>	<i>I</i>	<i>N</i>	<i>R</i>
<i>F</i>	<i>A</i>	<i>F</i>	<i>G</i>	<i>X</i>
<i>A</i>	<i>F</i>	<i>A</i>	<i>G</i>	<i>G</i>
<i>G</i>	<i>X</i>	<i>X</i>	<i>G</i>	<i>D</i>
<i>D</i>	<i>F</i>	<i>X</i>	<i>X</i>	<i>G</i>
<i>F</i>	<i>G</i>	<i>D</i>	<i>G</i>	<i>A</i>
				<i>A</i>

Finally, the ciphertext is obtained by reading down the columns (omitting the labels) in order:

FAGDFAFXFGFAXXDGGGXGXGDGAA.

Decryption is easy, as long as you know the keyword. From the length of the keyword and the length of the ciphertext, the length of each column is determined. The letters are placed into columns, which are reordered to match the keyword. The original matrix is then used to recover the plaintext.

The initial matrix and the keyword were changed frequently, making cryptanalysis more difficult, since there was only a limited amount of ciphertext available for any combination. However, the system was successfully attacked by the French cryptanalyst Georges Painvin and the Bureau du Chiffre, who were able to decrypt a substantial number of messages.

Here is one technique that was used. Suppose two different ciphertexts intercepted at approximately the same time agree for the first several characters. A reasonable guess is that the two plaintexts agree for several words. That means that the top few entries of the columns for one are the same as for the other. Search through the ciphertexts and find other places where they agree. These possibly represent the beginnings of the columns. If this is correct, we know the column lengths. Divide the ciphertexts into columns using these lengths. For the first ciphertext, some columns will have one length and others will be one longer. The longer ones represent columns that should be near the beginning; the other columns should be near the end. Repeat for the second ciphertext. If a column is long for both ciphertexts, it is very near the beginning. If it is long for one ciphertext and not for the other, it goes in the middle. If it is short for both, it is near the end. At this point, try the various orderings of the columns, subject to these restrictions. Each ordering corresponds to a potential substitution cipher. Use frequency analysis to try to solve these. One should yield the plaintext, and the initial encryption matrix.

The letters *ADFGX* were chosen because their symbols in Morse code (· —, — · ·, · · —, — — ·, — · · —) were not easily confused. This was to avoid

transmission errors, and represents one of the early attempts to combine error correction with cryptography. Eventually, the *ADFGX* cipher was replaced by the *ADFGVX* cipher, which used a 6×6 initial matrix. This allowed all 26 letters plus 10 digits to be used.

For more on the cryptanalysis of the *ADFGX* cipher, see [Kahn].

2.7 Block Ciphers

In many of the aforementioned cryptosystems, changing one letter in the plaintext changes exactly one letter in the ciphertext. In the shift, affine, and substitution ciphers, a given letter in the ciphertext always comes from exactly one letter in the plaintext. This greatly facilitates finding the key using frequency analysis. In the Vigenère system, the use of blocks of letters, corresponding to the length of the key, made the frequency analysis more difficult, but still possible, since there was no interaction among the various letters in each block. Block ciphers avoid these problems by encrypting blocks of several letters or numbers simultaneously. A change of one character in a plaintext block should change potentially all the characters in the corresponding ciphertext block.

The Playfair cipher in Section 2.6 is a simple example of a block cipher, since it takes two-letter blocks and encrypts them to two-letter blocks. A change of one letter of a plaintext pair will always change at least one letter, and usually both letters, of the ciphertext pair. However, blocks of two letters are too small to be secure, and frequency analysis, for example, is usually successful.

Many of the modern cryptosystems that will be treated later in this book are block ciphers. For example, DES operates on blocks of 64 bits. AES uses blocks of 128 bits. RSA uses blocks several hundred bits long, depending on the modulus used. All of these block lengths are long enough to be secure against attacks such as frequency analysis.

The standard way of using a block cipher is to convert blocks of plaintext to blocks of ciphertext, independently and one at a time. This is called the electronic codebook (ECB) mode. However, there are ways to use feedback from the blocks of ciphertext in the encryption of subsequent blocks of plaintext. This leads to the cipher block chaining (CBC) mode and cipher feedback (CFB) mode of operation. These are discussed in Section 4.5.

In this section, we discuss the Hill cipher, which is a block cipher invented in 1929 by Lester Hill. It seems never to have been used much in practice. Its significance is that it was perhaps the first time that algebraic methods (linear algebra, modular arithmetic) were used in cryptography in an essential way. As we'll see in later chapters, algebraic methods now occupy a central position in the subject.

Choose an integer n , for example $n = 3$. The key is an $n \times n$ matrix M whose entries are integers mod 26. For example, let

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 11 & 9 & 8 \end{pmatrix}.$$

The message is written as a series of row vectors. For example, if the message is *abc*, we change this to the single row vector $(0, 1, 2)$. To encrypt, multiply the vector by the matrix (traditionally, the matrix appears on the right in the multiplication; multiplying on the left would yield a similar theory) and reduce mod 26:

$$(0, 1, 2) \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 11 & 9 & 8 \end{pmatrix} \equiv (0, 23, 22) \pmod{26}.$$

Therefore, the ciphertext is *AXW*. (The fact that the first letter *a* remained unchanged is a random occurrence; it is not a defect of the method.)

In order to decrypt, we need the determinant of M to satisfy

$$\gcd(\det(M), 26) = 1.$$

This means that there is a matrix N with integer entries such that $MN \equiv I \pmod{26}$, where I is the $n \times n$ identity matrix.

In our example, $\det(M) = -3$. The inverse of M is

$$-\frac{1}{3} \begin{pmatrix} -14 & 11 & -3 \\ 34 & -25 & 6 \\ -19 & 13 & -3 \end{pmatrix}.$$

Since 17 is the inverse of $-3 \pmod{26}$, we replace $-1/3$ by 17 and reduce mod 26 to obtain

$$N = \begin{pmatrix} 22 & 5 & 1 \\ 6 & 17 & 24 \\ 15 & 13 & 1 \end{pmatrix}.$$

The reader can check that $MN \equiv I \pmod{26}$.

For more on finding inverses of matrices mod n , see Section 3.8.

The decryption is accomplished by multiplying by N , as follows:

$$(0, 23, 22) \begin{pmatrix} 22 & 5 & 1 \\ 6 & 17 & 24 \\ 15 & 13 & 1 \end{pmatrix} \equiv (0, 1, 2) \pmod{26}.$$